

Report No.: 244544885a 001 Page 1 of 16

Client: SHANGHAI NNW NEW MATERIALS TECHNOLOGY CO., LTD.

Contact Information: ROOM 402, Building 17, Lane 268, Lingxin Road, Changning District
Shanghai, CHINA

**Identification/
Model No(s):** Permanent Marker Ink
Black/Red/Blue/Green

Sample obtaining method: Sending by customer

Condition at delivery: Test item complete and undamaged.

Sample Receiving date: 2023-09-14

Testing Period: 2023-09-14 to 2023-09-22

Place of testing:

Test Specification:

1. Safety Data Sheet (SDS) ^

Test result:

Please refer to result
page

Other information:

* This report is only valid with the provided ingredient composition.

^ The test was subcontracted to other labs.

For and on behalf of
TÜV Rheinland (Shanghai) Co., Ltd.



2023-09-26

Neo Yang / Assistant Manager

Date

Name/Position

Sample information is provided by customer. Test result is drawn according to the kind and extent of tests performed.

This test report relates to the above mentioned test sample. Without permission of the test center this test report is not permitted to be duplicated in extracts. This test report does not entitle to carry any safety mark on this or similar products.

"Decision Rule" document announced in our website (<https://www.tuv.com/landingpage/en/qm-gcn/>) describes the statement of conformity and its rule of enforcement for test results are applicable throughout this test report.

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

• 1.1 Product identifier

• Trade name:

Permanent Marker Ink

• Other means of identification:

Registration number: No available data.

UFI code: No available data.

• 1.2 Relevant identified uses of the substance or mixture and uses advised against on

• Application of the substance/ mixture:

Consumer uses, writing.

• Uses advised against:

No available data.

• Reason why uses advised against:

No available data.

• 1.3 Details of the supplier of the safety data sheet

• Manufacturer/Supplier:

Shanghai NNW New Materials Technology Co., Ltd.

ROOM 402, Building17, Lane 268, Lingxin Road, Changning District Shanghai, CHINA

Post code: 200335

Tel: +086-21-64476059

Fax: +086-21-64476096

Email: sales@nnwchina.com

• Only Representative/other EU contact point:

No information available.

• Further information obtainable from:

Shanghai NNW New Materials Technology Co., Ltd.

• 1.4 Emergency telephone number

Tel: +8613311812200 (24 hours available)

SECTION 2: Hazards identification

• 2.1 Classification of the substance or mixture

• 2.1.1 Classification according to regulation (EC) 1272/2008:

Regulation (EC) No 1272/2008 [CLP]	Classification procedure
Flam. Liq. 3, H226	Calculation method
Eye Irrit. 2, H319	Additivity approach

• 2.1.2 Additional information:

For full text of Hazard- and EU Hazard-statements: see SECTION 16.

• 2.2 Label elements

• Labeling according to Regulation (EC) No 1272/2008:

The product is labelled according to CLP Regulation.

• Hazard pictograms:

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

GHS02 GHS07

• Signal word:

Warning

• Hazard-determining components of labelling:

Not applicable

• Hazard statements:

H226 Flammable liquid and vapour

H319 Causes serious eye irritation

• Precautionary statement:

P101 If medical advice is needed, have product container or label at hand.

P102 Keep out of reach of children.

P103 Read carefully and follow all instructions.

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P370 + P378 In case of fire: Use CO₂, chemical powder, water spray or alcohol resistant foam to extinguish. Do not use water with full jet

P403 + P235 Store in a well-ventilated place. Keep cool.

P501 Dispose of contents/container in accordance with local/regional/ national/international regulations.

• Supplemental label elements:

Not applicable

• 2.3 Other hazardsNone of the ingredients ($\geq 0.1\%$) meets the criteria for PBT/vPvB in accordance with Annex XIII.None of the ingredients ($\geq 0.1\%$) is identified as having endocrine disrupting properties according to Regulation (EU) 2017/2100.

SECTION 3: Composition/information on ingredients

• 3.1 Substance

Not applicable

• 3.2 Mixtures**• Description of the mixture:**

Permanent Marker Ink.

Ingredients:

Substance	CAS No.	Index No.	EC No.	w/w, %	CLP Classification	SCL/M-factor/ATE
Black						
Polymerized rosin	65997-05-9	-	500-163-2	35	None	-
Glycerin	56-81-5	-	200-289-5	18	None	-
Ethyl alcohol	64-17-5	603-002-00-5	200-578-6	17	Flam. Liq. 2, H225 Eye Irrit. 2, H319	Eye Irrit. 2, H319: C $\geq 50\%$
Methyl proxitol	107-98-2	603-064-00-3	203-539-1	14	Flam. Liq. 3, H226 STOT SE 3, H336	-

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

Propylene glycol	57-55-6	-	200-338-0	8	None	-
Solvent Black 5	11099-03-9	-	-	8	Skin Irrit.2, H315 Eye Irrit.2, H319	-
Red						
Polymerized rosin	65997-05-9	-	500-163-2	35	None	-
Glycerin	56-81-5	-	200-289-5	18	None	-
Ethyl alcohol	64-17-5	603-002-00-5	200-578-6	17	Flam. Liq. 2, H225 Eye Irrit. 2, H319	Eye Irrit. 2, H319: C _≥ 50%
Methyl proxitol	107-98-2	603-064-00-3	203-539-1	14	Flam. Liq. 3, H226 STOT SE 3, H336	-
Propylene glycol	57-55-6	-	200-338-0	8	None	-
Solvent Red 122	12227-55-3	-	-	8	None	-
Blue						
Polymerized rosin	65997-05-9	-	500-163-2	35	None	-
Glycerin	56-81-5	-	200-289-5	18	None	-
Ethyl alcohol	64-17-5	603-002-00-5	200-578-6	17	Flam. Liq. 2, H225 Eye Irrit. 2, H319	Eye Irrit. 2, H319: C _≥ 50%
Methyl proxitol	107-98-2	603-064-00-3	203-539-1	14	Flam. Liq. 3, H226 STOT SE 3, H336	-
Propylene glycol	57-55-6	-	200-338-0	8	None	-
Acid Blue 9	2650-18-2	-	220-168-0	8	None	-
Green						
Polymerized rosin	65997-05-9	-	500-163-2	35	None	-
Glycerin	56-81-5	-	200-289-5	18	None	-
Ethyl alcohol	64-17-5	603-002-00-5	200-578-6	17	Flam. Liq. 2, H225 Eye Irrit. 2, H319	Eye Irrit. 2, H319: C _≥ 50%
Methyl proxitol	107-98-2	603-064-00-3	203-539-1	14	Flam. Liq. 3, H226 STOT SE 3, H336	-
Propylene glycol	57-55-6	-	200-338-0	8	None	-
Solvent Blue 70	12237-24-0	-	-	4	Aquatic Chronic 3, H412	-
Solvent Yellow 79	12237-31-9	-	-	4	None	-

Additional information:

For full text of H-statements, see SECTION 16.

SECTION 4: First aid measures**4.1 Description of first aid measures****General advice:**

Call a POISON CENTER/doctor, if you feel unwell.

If medical advice is needed, have product container or label at hand.

After inhalation:

Remove person to fresh air and keep comfortable for breathing.

Call a POISON CENTER/doctor, if you feel unwell.

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink**After skin contact:**

Take off immediately all contaminated clothing.

Rinse skin with water/ shower.

If there are signs of irritation or other symptoms seek medical attention.

After eye contact:

Rinse cautiously with water for several minutes.

Remove contact lenses, if present and easy to do. Continue rinsing.

If eye irritation persists: Get medical advice/attention.

After swallowing:

Rinse mouth.

Call a POISON CENTER/doctor if you feel unwell.

• 4.2 Most important symptoms and effects, both acute and delayed:

Acute and delayed effects are as indicated in sections 2 and 11.

• 4.3 Indication of any immediate medical attention and special treatment needed:

Treatment according to symptoms, no known specific medicine.

SECTION 5: Fire-fighting measures**• 5.1 Extinguishing media****• Suitable extinguishing agents:**

Use CO₂, chemical powder, water spray or alcohol resistant foam to extinguish.

• Unsuitable extinguishing media:

Water with full jet.

• 5.2 Special hazards arising from the substance or mixture:

Flammable liquid.

Vapors may form explosive mixture with air.

• 5.3 Advice for firefighters**Protective equipment:**

Wear self-contained breathing apparatus for firefighting.

SECTION 6: Accidental release measures**• 6.1 Personal precautions, protective equipment and emergency procedures****• 6.1.1 For non-emergency personnel****Protective equipment:**

Protective gloves/ eye protection/ face protection/ dust mask.

Emergency procedures:

Ensure well ventilation.

Eliminate all ignition sources (no smoking, flares, sparks or flames in immediate area).

Wear personal protective equipment.

Avoid breathing vapor.

Beware of accumulation of vapor in low areas or contained areas, where explosive concentrations may occur.

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

Avoid contact with eyes.

• 6.1.2 For emergency responders**Personal protective equipment:**

Protective gloves/ eye protection/ face protection/ dust mask.

• 6.2 Environmental precautions:

Prevent further leakage or spillage if safe to do so.

Prevent spillage from entering drains, sewer, basement or confined areas.

If the spillage contaminates rivers, lakes or drains inform respective authorities.

• 6.3 Methods and material for containment and cleaning up:

Absorb with liquid-binding material (sand, diatomite, acid binders, universal binders, sawdust).

Dispose contaminated material as waste according to section 13.

• 6.4 Reference to other sections:

See section 7 for information on safe handling.

See section 8 for information on personal protection equipment.

See section 13 for disposal information.

SECTION 7: Handling and storage**• 7.1 Precautions for safe handling:**

Read carefully and follow all instructions.

Use only in well ventilated areas.

Avoid all sources of ignition.

Avoid breathing vapors.

Use respiratory protective device against the effects of vapor.

Avoid contact with eyes.

• Information about fire and explosion protection:

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources.

• 7.2 Conditions for safe storage, including any non-compatibility**• Requirements to be met by storerooms and receptacles:**

Store in a cool and well-ventilated place.

Keep container tightly closed.

• Information about storage in one common storage facility:

Keep away from ignition source.

• Further information about storage conditions:

Store locked up.

• Storage class:

3.

• 7.3 Specific end use(s):

See section 1.2.

SECTION 8: Exposure controls/personal protection

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink**• 8.1 Control parameters****• Ingredients with limit values that require monitoring at the workplace:**

Country	Limit value - Eight hours	Limit value – Short term
56-81-5 Glycerin		
Belgium	10 mg/m ³	-
Finland	20 mg/m ³	-
France	10 mg/m ³	-
Germany (DFG)	200 mg/m ³ Inhalable fraction	400 mg/m ³ Inhalable fraction; 15 minutes average value
Ireland	10 mg/m ³	-
Poland	10 mg/m ³	-
Spain	10 mg/m ³	-
United Kingdom	10 mg/m ³	-
64-17-5 Ethyl alcohol		
Austria	1000ppm; 1900 mg/m ³	2000ppm; 3800 mg/m ³
Belgium	1000ppm; 1907 mg/m ³	-
Denmark	1000ppm; 1900 mg/m ³	2000ppm; 3800 mg/m ³
Finland	1000ppm; 1900 mg/m ³	1300ppm; 2500 mg/m ³ 15 minutes average value
France	1000ppm; 1900 mg/m ³	5000ppm; 9500 mg/m ³
Germany (AGS)	500ppm; 960 mg/m ³	1000ppm; 1920mg/m ³ 15 minutes average value
Germany (DFG)	500ppm; 960 mg/m ³	1000ppm; 1920mg/m ³ 15 minutes average value
Hungary	1900 mg/m ³	7600 mg/m ³
Ireland	-	1000ppm 15 minutes average value
Latvia	1000 mg/m ³	-
Poland	1900 mg/m ³	-
Spain	-	1000ppm; 1910 mg/m ³
Sweden	500ppm; 1000 mg/m ³	1000ppm; 1900mg/m ³ 15 minutes average value
The Netherlands	260 mg/m ³	1900 mg/m ³
United Kingdom	1000ppm; 1920 mg/m ³	-
107-98-2 Methyl proxitol		
Austria	50ppm; 187mg/m ³	50ppm; 187mg/m ³
Belgium	100ppm; 375mg/m ³	150ppm; 568mg/m ³
Denmark	50ppm; 185mg/m ³	100ppm; 370mg/m ³
European Union	100ppm; 375mg/m ³	150ppm; 568mg/m ³
Finland	100ppm; 370mg/m ³	150ppm; 560mg/m ³
France	50ppm; 188mg/m ³	100ppm; 375mg/m ³
Germany (AGS)	100ppm; 370mg/m ³	200ppm; 740mg/m ³ 15 minutes average value
Germany (DFG)	100ppm; 370mg/m ³	200ppm; 740mg/m ³
Hungary	375mg/m ³	568mg/m ³
Ireland	100ppm; 375mg/m ³	150ppm; 568mg/m ³ 15 minutes reference period
Italy	100ppm; 375mg/m ³	150ppm; 568mg/m ³
Latvia	100ppm; 375mg/m ³	150ppm; 568mg/m ³ 15 minutes average value
Romania	100ppm; 375mg/m ³	150ppm; 568mg/m ³ 15 minutes average value
Spain	100ppm; 375mg/m ³	150ppm; 568mg/m ³
Sweden	50ppm; 190mg/m ³	150ppm; 568mg/m ³ 15 minutes average value

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

United Kingdom	100ppm; 370mg/m ³	150ppm; 560mg/m ³
57-55-6 Propylene glycol		
Ireland	150 ppm; 470 mg/m ³	-
United Kingdom	150 ppm; 474 mg/m ³	-

• DNELs:

DNEL type		DNEL worker value	DNEL consumer value
64-17-5 Ethyl alcohol			
Systemic effects	Long-term, inhalation exposure	380 mg/m ³	114 mg/m ³
	Long-term, dermal exposure	343 mg/kg bw/day	206 mg/kg bw/day
	Long-term, oral exposure	-	87 mg/kg bw/day
Local effects	Acute /short term, inhalation exposure	1 900 mg/m ³	950 mg/m ³
107-98-2 Methyl proxitol			
Systemic effects	Long-term, inhalation exposure	369 mg/m ³	43.9 mg/m ³
	Acute /short term, inhalation exposure	553.5 mg/m ³	-
	Long-term, dermal exposure	183 mg/kg bw/day	78 mg/kg bw/day
	Long-term, oral exposure	-	33 mg/kg bw/day
Local effects	Acute /short term, inhalation exposure	553.5 mg/m ³	-

• PNECs:

PNEC type	Value
64-17-5 Ethyl alcohol	
Freshwater	960 µg/L
Intermittent releases (freshwater)	2.75 mg/L
Marine water	790 µg/L
Sewage treatment plant (STP)	580 mg/L
Sediment (freshwater)	3.6 mg/kg sediment dw
Sediment (marine water)	2.9 mg/kg sediment dw
Soil	630 µg/kg soil dw
Secondary poisoning	380 - 720 mg/kg food
107-98-2 Methyl proxitol	
Freshwater	10 mg/L
Intermittent releases (freshwater)	100 mg/L
Marine water	1 mg/L
Sewage treatment plant (STP)	100 mg/L
Sediment (freshwater)	52.3 mg/kg sediment dw
Sediment (marine water)	5.2 mg/kg sediment dw
Soil	4.59 mg/kg soil dw

• Additional information:

The lists valid during the marking were used as basis.

• 8.2 Exposure controls

• Based on the composition shown in section 3, the following measures are suggested for occupational safety measure.

• Appropriate engineering controls:

Handle in accordance with good industrial hygiene and safety practice.

Wash hands and face before breaks and at the end of work.

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

Take off contaminated clothing and wash it before reuse.

See section 7 for information about design of technical facilities.

• Personal protective equipment**• Eye and face protection:**

Not required under normal conditions of use.

For industrial applications, following protection required:

**Safety glasses**

Protective goggles with side-shields.

• Skin protection**• Hand protection:**

Not required under normal conditions of use.

For industrial applications, following protection required:

**Protective gloves**

Gloves made from butyl rubber Neoprene™ rubber, nitrile rubber (thickness > 0.3mm; breakthrough times up to 480 minutes).

• Other skin protection:

Not required under normal conditions of use.

For industrial applications, following protection required:

**Face protection****Safety boots****Safety overalls**

Face shield, gauntlets, boots, bodysuit are recommended.

• Respiration protection:

Not required under normal conditions of use.

For industrial applications, following protection required:

**Respiratory equipment**

Use positive pressure breathing mask if concentrations in air could exceed occupational exposure standard.

• Thermal hazards:

Not required under normal conditions of use.

For industrial applications, following protection required:

The gauntlets, boots, bodysuit and other personal protective equipment must be flame retardant and no heat-conducting.

• Environmental exposure controls:

Control measures must be made in accordance with Community environmental protection legislation.

SECTION 9: Physical and chemical properties

• 9.1 Information on basic physical and chemical properties**• Physical state**

Liquid

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

• Color	Black/Red/Blue/Green
• Odor & Odor threshold	Odorless
• Melting point/freezing point (or softening point/range)	Not determined
• Boiling point or initial boiling point and boiling range	78°C (Ethyl alcohol)
• Flammability	Highly flammable liquid
• Lower and upper explosion limit	Lower explosion limit: 3.1 vol.% 59 g/m ³ (Ethyl alcohol) Upper explosion limit: 27.7 vol.% 532 g/m ³ (Ethyl alcohol)
• Flash point	29°C (calculated value)
• Auto-ignition temperature	400°C (Ethyl alcohol)
• Decomposition temperature	>700°C (Ethyl alcohol)
• pH	4-8 (20°C, 100g/l)
• Kinematic viscosity (mm ² /s)	Not determined
• Solubility	Not determined
• Partition coefficient n-octanol/water (log value)	Not applicable
• Vapor pressure	Not determined
• Density and/or relative density	Not determined
• Relative vapor density	Not determined
• Particle characteristics	Not determined
• 9.2 Other information	
• 9.2.1 Information with regard to physical hazard classes:	
• Explosives	Not applicable
• Flammable gases	Not applicable
• Aerosols	Not applicable
• Oxidising gases	Not applicable
• Gases under pressure	Not applicable
• Flammable liquids	Not determined
• Flammable solids	Not applicable
• Self-reactive substances and mixtures	Not applicable
• Pyrophoric liquids	Not applicable
• Pyrophoric solids	Not applicable
• Self-heating substances and mixtures	Not applicable
• Substances and mixtures, which emit flammable gases in contact with water	Not applicable
• Oxidizing liquids	Not applicable
• Oxidizing solids	Not applicable
• Organic peroxides	Not applicable
• Corrosive to metals	Not applicable
• Desensitised explosives	Not applicable
• 9.2.2 Other safety characteristics	
• Mechanical sensitivity	Not applicable
• Self-accelerating polymerisation temperature	Not applicable
• Formation of explosible dust/air mixtures	Not applicable
• Acid/alkaline reserve	Not applicable

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

• Evaporation rate	Not determined
• Miscibility	Not determined
• Conductivity	Not applicable
• Corrosiveness	Not applicable
• Gas group	Not applicable
• Redox potential	Not applicable
• Radical formation potential	Not applicable
• Photocatalytic properties	Not applicable
• Other physical and chemical parameters	Not determined

SECTION 10: Stability and reactivity

• 10.1 Reactivity:

The product is non-reactive under normal conditions of use, storage and transport.

• 10.2 Chemical stability:

Under storage at normal ambient temperatures, the product is stable.

• 10.3 Possibility of hazardous reactions:

No known hazardous reaction.

• 10.4 Conditions to avoid:

Heat/sparks/open flames/hot surfaces.

• 10.5 Incompatible materials:

Strong oxidizing agent and acids.

• 10.6 Hazardous decomposition products:

Does not decompose when used for intended uses.

SECTION 11: Toxicological information

• 11.1 Information on toxicological effects

• Acute toxicity:

Based on available data, the classification criteria are not met.

• LD50/LC50 values relevant for classification:

64-17-5 Ethyl alcohol

Rat	LD50-oral	7060mg/kg
	LC50-inhalation	20000ppm/10H
Rabbit	LD50-oral	6300 mg/kg
Guinea pig	LD50-oral	5560mg/kg
Mouse	LD50-oral	3450 mg/kg
	LC50-inhalation	39000 mg/m3 /4H

107-98-2 Methyl proxitol

Mouse	LD50-oral	11700mg/kg
Rabbit	LD50-skin	13000mg/kg

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

	LD50-oral	5700mg/kg
Rat	LC50-inhalation	10000ppm/5H
Remark: All the above data are from literature.		

• **Skin corrosion/irritation:**

Based on available data, the classification criteria are not met.

• **Serious eyes damage/ irritation:**

Causes serious eye irritation.

• **Respiratory or skin sensitization:**

Based on available data, the classification criteria are not met.

• **Germ cell mutagenicity:**

Based on available data, the classification criteria are not met.

• **Carcinogenicity:**

Based on available data, the classification criteria are not met.

• **Reproductive toxicity:**

Based on available data, the classification criteria are not met.

• **STOT-single exposure:**

Based on available data, the classification criteria are not met.

• **STOT-repeated exposure:**

Based on available data, the classification criteria are not met.

• **Aspiration hazard:**

Based on available data, the classification criteria are not met.

• **11.2 Information on other hazards**

• **11.2.1 Endocrine disrupting properties:**

None of the ingredients ($\geq 0.1\%$) is considered to have endocrine-disrupting properties with respect to humans, as none of them meet the criteria set out in section A of Regulation (EU) No 2017/2100.

• **11.2.2 Other information:**

No known other relevant information on adverse health effects.

SECTION 12: Ecological information

• **12.1 Toxicity:**

LC50/EC50/NOEC values relevant for classification:

64-17-5 Ethyl alcohol

Short-term toxicity to fish

LC50 (4 days) 14.2 - 15.4 g/L

EC50 (4 days) 12.7 - 12.9 g/L

Long-term toxicity to fish

NOEC (5 days) 250 - 1 000 mg/L

Short-term toxicity to aquatic invertebrates

EC50 (48 h) 10 g/L

LC50 (48 h) 5.012 g/L

Long-term toxicity to aquatic invertebrates

NOEC (10 days) 2 - 9.6 mg/L

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

	LC50 (10 days) 1.806 g/L LC50 (48 h) 9.248 g/L
Toxicity to aquatic algae and cyanobacteria	EC50 (4 days) 675 - 22 000 mg/L
Toxicity to aquatic plants other than algae	EC50 (7 days) 4.432 - 5.967 g/L
107-98-2 Methyl proxitol	
Short-term toxicity to fish	LC50 (4 days) 6.812 g/L
Short-term toxicity to aquatic invertebrates	LC50 (48 h) 23.3 g/L
Toxicity to aquatic algae and cyanobacteria	EC50 (7 days) 1 g/L

• 12.2 Persistence and degradability:

64-17-5	Ethyl alcohol	BOD ₅ =1.067 - 1.236; COD=1.99; Readily biodegradable in water
107-98-2	Methyl proxitol	Readily biodegradable in water

• 12.3 Bio-accumulative potential:

64-17-5	Ethyl alcohol	Log Pow = -0.77 - 0.3
107-98-2	Methyl proxitol	Log Pow = 6.5

• 12.4 Mobility in soil:

64-17-5	Ethyl alcohol	Dissociation constant=15.9
---------	---------------	----------------------------

• 12.5 Results of PBT and vPvB assessment:

This mixture does not contain any substances ($\geq 0.1\%$) that are assessed to be a PBT or a vPvB.

• 12.6 Endocrine disrupting properties:

None of the ingredients ($\geq 0.1\%$) is considered to have endocrine-disrupting properties with respect to non-target organisms, as none of them meet the criteria set out in section B of Regulation (EU) No 2017/2100.

• 12.7 Other adverse effects:

No known other adverse effects.

• 12.8 Additional ecological information**• General notes:**

WGK1 (German Regulation) (self-assessment): slightly hazardous to water.

Do not allow large quantities of the product to reach ground water, water course or sewage system.

SECTION 13: Disposal consideration

• 13.1 Waste treatment methods**• Recommendation:**

Must not be disposed together with household garbage.

• 13.2 Un-cleaned packaging**• Recommendation:**

Dispose of contents/container in according to the local/regional/national/ international regulation.

SECTION 14: Transport information

• ADR/ADN/RID (2023 Edition)

This product is not subject to ADR/ADN according to special provision 216.

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

• IMO/IMDG (40-20)	This product is not subject to IMO/IMDG code according to special provision 216.
• IATA-DGR (64th)	This product is not subject to IATA-DGR according to special provision A46.

SECTION 15: Regulatory information

• **15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture**

• **MAK (German Maximum Workplace Concentration):**

64-17-5	Ethyl alcohol	5
---------	---------------	---

• **Directive 2012/18/EU**

• **Named dangerous substances-ANNEX I:**

None of the ingredients is listed.

• **Seveso category:**

P5c FLAMMABLE LIQUIDS

• **Qualifying quantity (tonnes) for the application of lower-tier requirements:**

5000 ton(net).

• **Qualifying quantity (tonnes) for the application of upper-tier requirements:**

50000 ton(net).

• **National regulations.**

• **Water hazard class:**

WGK1 (German Regulation) (self-assessment): slightly hazardous to water.

• **Other regulations, limitations and prohibitive regulations**

• **SVHC Candidate list of REACH Regulation Annex XIV Authorization:**

None of the ingredients is listed.

• **REACH Regulation Annex XVII Restriction:**

None of the ingredients is listed.

• **REACH Regulation Annex XIV Authorization List:**

None of the ingredients is listed.

• **15.2 Chemical safety assessment:**

A Chemical Safe Assessment has not been carried out.

SECTION 16: Other information

• **16.1 Indication of changes:**

None.

• **16.2 Abbreviations and acronyms:**

ADR: Accord européen sur le transport des marchandises dangereuses par Route (European Agreement concerning the International Carriage of Dangerous Goods by Road).

IMDG: International Maritime Code for Dangerous Goods.

IATA: International Air Transport Association.

CAS: Chemical Abstracts Service (division of the American Chemical Society)

DNEL: Derived No-Effect Level (REACH)

PNEC: Predicted No-Effect Concentration (REACH)

PBT: Persistent, Bio accumulative and Toxic

vPvB: very persistent and very bio accumulative

Safety Data Sheet

Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

SVHC: Substance of Very High Concern

LD50: Lethal dose, 50 percent

LC50: Lethal concentration, 50 percent

EC50: Concentration of maximal effect, 50 percent

NOEC: No observed effect concentration

Flam. Liq. 2: Flammable liquid, hazard category 2

Flam. Liq. 3: Flammable liquids, hazard category 3

Skin Irrit. 2: Skin corrosion/irritation, hazard category 2

Eye Irrit. 2: Eye damage/irritation, hazard category 2

STOT SE 3: Specific target organ toxicity after single exposure, hazard category 3

Aquatic Chronic 3: Long-term (chronic) aquatic hazard, hazard category 3

• 16.3 Key literature references and sources for data:<https://echa.europa.eu/><https://chem.nlm.nih.gov/><https://www.osha.gov/><http://www.unece.org/><http://www.imo.org/><https://www.dguv.de/><https://epa.govt.nz/><http://www.ilo.org/><https://www.phmsa.dot.gov/>**• 16.4 Classification for mixtures and used evaluation method according to regulation (EC) 1272/2008 [CLP]:**

See section 2.1 (classification).

• 16.5 Relevant H- and EUH-phrases (number and full text):

H225 Highly flammable liquid and vapour

H226 Flammable liquid and vapour

H315 Causes skin irritation

H319 Causes serious eye irritation

H336 May cause drowsiness or dizziness

H412 Harmful to aquatic life with long lasting effects

• 16.6 Training advice:

Workers must be trained in general and specific aspects of safe work procedures for this chemical. Workers need easy to understand guidelines and instructions for safe working with this chemical. Instructions, guidelines and legally required documents need to be available for workers in easy to understand language.

• 16.7 Further information:

The contents and format of this SDS are in accordance with Regulation (EC) No 1907/2006, its amendment Regulation (EU) No 2020/878 and (EC) No 1272/2008.

DISCLAIMER OF LIABILITY:

The information in this SDS was obtained from sources which we believe are reliable. However, the information is provided without any warranty, express or implied, regarding its correctness. The conditions or methods of handling, storage, use or disposal of the product are beyond our control and may be beyond our knowledge. For this and other reasons, we do not assume responsibility and expressly disclaim liability for loss, damage or expense arising out of or in any way connected with the handling, storage, use or

Safety Data Sheet
Regulation (EC) No. 1907/2006 and 1272/2008

Printing Date: September 25, 2023

Version No.: 1.0

Revision: September 25, 2023

Trade Name: Permanent Marker Ink

disposal of the product. This SDS was prepared and is to be used only for this product. If the product is used as a component in another product, this SDS information may not be applicable.

End of safety data sheet

